

Sentinel — Project Status Report

Automated strategic-intelligence agent · Adapting to the new capability specification

Yassine Dahmoul · Intern · GitLab branch `sentinel-agent-yassine`

Where we stand. The original Sentinel MVP is **built, deployed and running autonomously**. The team has since specified two new capabilities — a parameterized ~3000-word deep-research report and a competitor-comparison report — which require the agent to become *configurable* rather than single-topic. Three foundation phases are now complete: the system is fully parameterized (any theme, language, geography) while the production weekly report remains **byte-for-byte unchanged**. The next phase resolves the one open technical risk before the new reports are built.

v1 live

weekly agent running

3 / 8

adaptation phases done

252

automated tests passing

0 €

running cost

1. What we originally built

Sentinel v1 is a **weekly autonomous pipeline** watching the AI SaaS B2B market. It runs unattended in the cloud every Monday morning and emails a sourced intelligence report.

1 · Collect

2 · Process

3 · Full text

4 · Analyze

5 · Trends

6 · Report

7 · Deliver

CAPABILITY	HOW IT WORKS
Collection	5 free sources — RSS feeds, Hacker News, Product Hunt, Google News, GNews. ~2,700 articles per run.
Filtering	Relevance filter + two-layer deduplication (exact URL, then cross-source title similarity). ~2,700 → ~1,700.
AI analysis	Google Gemini summarizes and tags each article against a fixed 20-topic taxonomy; Groq as fallback.
Historical memory	Supabase database persists between runs, so the agent distinguishes a <i>new</i> trend from an <i>accelerating</i> one.
Report & delivery	HTML report in 5 sections, emailed via Gmail + optional Slack notification.
Automation	GitHub Actions weekly cron (UTC-timed for Monday morning CET) + a keep-alive job preventing the free database from pausing.

Two design principles worth knowing. (1) **It is a workflow, not an autonomous agent** — the steps run in a fixed order and the LLM only acts inside two narrow, controlled steps, which makes it predictable and debuggable. (2) **The database is the memory** — cloud runners are wiped after every run, so all history lives in Supabase; that is what makes trend detection possible at all.

2. Why the system had to change

The team's new specification asks the user to fill in a **request form** with six variables, and to receive a much deeper report on a schedule:

VARIABLE	MEANING	EXAMPLE
{A}	Theme of the watch	"AI in healthcare"
{B}	Report language	French
{C}	Geographic zone	France / Europe / World
{D}	Time horizon	last 12 months + 3–5 year projection
{E}	Sector focus	health, retail, education...
{F}	Executive priority	growth, innovation, M&A...

Comparing that to what existed revealed **three structural gaps**:

Gap 1 — Nothing was parameterizable

The theme was not a setting: "AI SaaS B2B" was written directly into the code in about **11 separate places** — prompts, the Google News URL, report headings, even the word "Welyne" inside a prompt. Changing the topic was not a configuration change; it was a code change.

Gap 2 — The report is ~18% of the required length

Today's report contains roughly **548 words** of AI-written narrative. The specification asks for **3,000 ± 250**, in a mandatory three-part structure with scenarios and weak signals. This is not a longer prompt — it needs a different generation method.

Gap 3 — The anti-hallucination rule blocked the new requirement

Sentinel enforces a strict rule: *every claim must cite a collected source article; anything uncited is deleted*. That is precisely why its reports are trustworthy. But the new spec requires citing **McKinsey, BCG, Gartner, IDC** and producing **3–5 year projections** — neither of which can cite last week's news article. The existing code would have silently deleted every such statement. The requirement was not merely unimplemented; it was **structurally impossible**.

3. What we have implemented so far

Three phases of foundation work, deliberately **additive and invisible**: the production weekly report behaves exactly as before, while the machinery underneath became configurable.

Foundations & safety net Done

- **Regression safety net.** A test renders the weekly report and compares it byte-for-byte against a reference copy. We validated the net itself by deliberately breaking a heading — the test caught it immediately. Every later change relied on this.
- **Reporting periods** generalized from weekly-only to weekly / monthly / quarterly.
- **LLM controls** added (output limits, tool support, truncation detection) — all optional, so current behaviour is unchanged.
- **Database** extended with report type, language, period, word count, and a new **citation ledger** table making every cited source auditable after the fact.

Full parameterization Done

The system is no longer single-topic. A request is now a small YAML file carrying {A}–{F}; it is overlaid onto the configuration and flows through the pipeline — **without a single pipeline stage being modified**.

Verified working (live run, French healthcare request):

- French-language queries drove collection (525 articles)
- A different keyword set filtered differently
- **Gemini produced the analysis in French**
- Report headings rendered in French

Verified nothing broke:

- No request profile ⇒ **literally zero change**
- Default search URL asserted **identical** to the old one
- English report still **byte-identical**
- Test suite grew 148 → **252 passing**

4. What is still to come

STEP	DELIVERABLE	WHAT IT ADDS	STATUS
1	Web search & citation policy	Real web search so the report can cite McKinsey / Gartner / IDC, plus the rewritten three-tier citation rule that unblocks projections.	Decisive
2	Deep-research report	The ~3,000-word deliverable: operational synthesis, detailed analysis with weak signals and scenarios, opportunities & risks.	Planned
3	Competitor report	Capability 2: 5–10 competitors ranked by threat, full dossiers, top-3 synthesis and action plan, driven by a company profile.	Planned
4	Periodic scheduling	Per-request cadence (weekly / monthly / quarterly) — the "programmer la réception périodique" part of the spec.	Planned
5	Multi-theme memory	Tracking several themes simultaneously, each with its own independent trend history.	Planned

The one open risk

Can we cite McKinsey and Gartner credibly, for free? The Google AI library we already use ships a **web-search capability that returns the URLs the model actually consulted** — requiring no new dependency and no paid service. What is not yet confirmed is whether it is included in the **free tier**. The next step is deliberately a short, cheap experiment to answer this *before* anything depends on it. If the answer is no, a fallback (sourcing from selected institutional RSS feeds) is already designed as a configuration switch rather than a rewrite.

How trustworthiness is preserved

The citation rule is being replaced by three enforced tiers rather than relaxed:

TIER	WHAT IT MEANS	HOW IT IS ENFORCED
A	Cites an article Sentinel collected	Today's rule, unchanged
B	Cites a web source Sentinel actually retrieved	Checked against the real URLs returned with the search
C	A projection or scenario	Allowed only if anchored to A/B evidence and visibly labelled as a hypothesis

Two safeguards make this robust: the AI **never writes a URL itself** (it references evidence identifiers, so invented links are impossible), and the publisher name is stored separately from the link, so attribution survives even if a link expires.